

Title:

How to Use 2g Servos with a 2S Battery (Without Burning Them Out!)

Description:

Many lightweight RC aircraft use tiny 2g servos, but these servos aren't designed to run directly from a 2S (7.4V) battery. In this video, I show a simple, inexpensive way to use 2g servos safely with a 2S power system without paying for expensive all-in-one receiver/ESC units.

I demonstrate the setup using the popular FlySky FS-i6 transmitter and AFHDS 2A receivers, although many of the concepts apply to other RC radio systems as well.

In this video you'll learn:• Why most 2g servos fail on 2S batteries.• The advantages and disadvantages of receivers with built-in ESCs and BECs.• Why a failed onboard ESC can limit future use of your receiver.• How I use a low-cost receiver, separate ESC and voltage regulator (BEC) to power 2g servos safely.• The wiring and setup details needed to make this budget solution work reliably.

If you're building lightweight foam RC aircraft or looking for an inexpensive 2S electronics setup, this approach can save money while protecting your servos.

If you found this video helpful, please Like, Subscribe, and leave a comment with your own experiences or questions.

Links:

Note, these prices (A\$) excluding postage will vary depending on vendor and time of year.

Flysky FS-i6 Transmitter - \$63.69+\$2.36 postage

FLYSKY FS-i6X i6X AFHDS 2A RC Transmitter with X6B iA6B iA10B Receiver 10CH 2.4GHz for Airplane Helicopter FPV Racing Drones

https://a.aliexpress.com/_mOYtmCF

4-Channel 1-2S Receiver: \$8.09

FS2A 4CH AFHDS 2A Mini Compatible Receiver PWM Output for Flysky i6 i6X i6S / FS-i6 FS-i6X FS-i6S Transmitter

https://a.aliexpress.com/_mNfHWsx

2S ESC: \$4.69

3.3V voltage regulator: 12A OPTO Brushless ESC Lightweight Electronic Speed Controller For FPV Racing Drone RC Airplane Multicopter 1104 1106 1806 Motor

https://a.aliexpress.com/_mPSunLd

Voltage Regulator: \$5.82 for 6x

Tiny Out 1A 1.2V 1.8V 2.5V 3.3V 5V 80mV Drop Voltage 60uA Standby Step-Down Buck LDO Module rep AMS1117 7805 for Arduino ESP32

https://a.aliexpress.com/_mMVEhvt

2g Servos: \$7.03

DSPOWER 2g Micro Digital Coreless Servo Plastic Gear JR/JST Plug Mini Servos for Rc Helicopter Plane Boat Car Model Accessoires

https://a.aliexpress.com/_mOzbonh

Choose one of 3 connectors:

JR2.54 (large), JST1.0 and JST1.25 - needs to match the sockets on your receiver.

For the FS2A receiver above, I used the JR2.54 connectors on my Jump Racer Jnr, which are very robust.

MC1104 Motor: \$9.19

1104 Mini Brushless Motor 14mm 4300kv for 1S-3S RC Drone FPV Racing Quadcopter

https://a.aliexpress.com/_mPIMTof

RS1102 Motor: \$53.69 for 4

4PCS Happymodel RS1102 1102 10000KV 13500KV 2S Brushless Motor 1.5mm Shaft for Mobula7 O4 Micro FPV Racing Drone RC Accessories

https://a.aliexpress.com/_mNSkedz

3016 Propellers: \$12.69 for 2x pairs (for the RS1102 motors)

2 pairs / 6 pairs Gemfan 3016 Propeller 1.5m hole 3 inch 3-Blade CW CCW FPV Propeller Mini
Props For 3inch FPV Racing Drone
https://a.aliexpress.com/_mKrvzWL

5040 Propellers: \$7.72 for 5x (for the MC1104 motors)
5PCS 5030 6035 7035 8040 8060 9050 1060 1160 Propeller For RC Airplane
https://a.aliexpress.com/_mrhoMXt

7A 2S/5A 3S Receiver with onboard ESC, BEC and Servo sockets:
\$38.79 Ma-RX62HE-A FLYSKY AFHDS 2A 6CH Micro Receiver Built-in 7A/2S(5A/3S) Brushless
ESC for FLYSKY Radio Transmitter
Select 1.5, 1.25, 1.5mm 3P connectors
https://a.aliexpress.com/_mrr6tTl
I've used this on my Chaotix 3D Flyer

2S Battery Example: \$3.95 on special and shipped from NSW, Australia
Turnigy nano-tech 300mAh 2S 35~70C Lipo Pack
https://hobbyking.com/en_us/turnigy-nano-tech-300mah-2s-35-70c-lipo-pack.html

#RCPlane #FlySky #FSi6 #AFHDS2A #2GServo #BEC #ESC #2SBattery #RCAircraft #FoamPlane
#DavesFunRC

See my video on YouTube: <https://youtu.be/1DB6VqkKScs>

0:00 Introduction
1:35 Choosing Radio Control Components
2:18 Choice of Options
2:23 Option 1: Integrated Receiver & ESC
2:43:Option 2: Separate Receiver, ESC and BEC
3:04 Examples of these options
3:57 2S ESC that might not work
4:12 Gotcha with 2S ESCs
4:34 Wiring Diagram for Option 2
5:02 Example Planes
5:26 Summing Up and Sign-off

Links

DMFV Jump Racer Build Video

<https://youtu.be/LZhj30esvAo>

DMFV Academy

[https://www.google.com/search?q=DMFV-Jump-Racer-Konstruktion&oq=DMFV-Jump-Racer-](https://www.google.com/search?q=DMFV-Jump-Racer-Konstruktion&oq=DMFV-Jump-Racer-Konstruktion&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCDEyMzRqMGo3qAIAAsAIA&sourceid=chrome&ie=UTF-8#fpstate=ive&vld=cid:d6b75a65,vid:Bi0jUbTTliU,st:0)

[Konstruktion&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCDEyMzRqMGo3qAIAAsAIA&sourceid=chrome&ie=UTF-8#fpstate=ive&vld=cid:d6b75a65,vid:Bi0jUbTTliU,st:0](https://www.google.com/search?q=DMFV-Jump-Racer-Konstruktion&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCDEyMzRqMGo3qAIAAsAIA&sourceid=chrome&ie=UTF-8#fpstate=ive&vld=cid:d6b75a65,vid:Bi0jUbTTliU,st:0)

ZART Shop in Melbourne for Printing Foam

ZART.COM.AU

Bambu P1P 3D Printer

<https://bambulab.com/en/p1?product=p1p>

Link to DMFV Jump Racer 1S Maiden Frights (Flights)

0:00 Introduction

0:41 Jump Racer Jnr Specs

1:38 Jump Racer Construction

3:44 Maiden Flights

6:55 Conclusions

7:06 Outro and Signos

Maiden Flights Script

Introduction

In my last video, I had fun building a 56cm wingspan DMFV Jump Racer using printing foam. I gave it a bright colour scheme and a jaunty pilot, so was very eager to see how she would fly...

I took the freshly built plane to Indoor Aviation at Mullum Mullum, full of anticipation. The build had gone well. I had also done some thrust tests and I thought that I would have enough oomph to get it airborne, using my new little brushless motor, running 1S.

Check out how the maiden flights went!

Test Flights

Well that didn't go great, didn't it? There were issues preventing the plane from staying airborne. In essence, the thrust was not enough for the weight and efficiency of this plane.

I tried 250,500 and 800mAh batteries.

I adjusted the ailerons to give some flap and I hand launched it.

The best I got was one length of the stadium, before I had to safely land it.

Diagnosis

After a phone consultation with the aeronautical engineer in the family, I concluded the following:

- The 5030 prop was out of balance - vibration is a thrust killer!
- The prop pitch was likely too fine for the revs that I get from 1S - like driving a car stuck in 1st gear
- Making the prop courser, gives less thrust in a static test, but will kick in once the plane is airborne
- The wing doesn't have an aerofoil shape and is symmetrical, which is good for inverted flight but not great for lift - it also has a very blunt leading edge
- The thrust to weight is about 1:2 so is a bit marginal with not a great lifting wing
- I had calibrated the brushless ESC a couple of times, so that was not the cause.

The Fix

- Balance the prop - I actually found out that all my 5030 props in the last batch I bought were out of balance and had extra weight on the opposite side of the 5030 marking.
- I have since balanced the DMFV prop as well as the 5030 prop that I run 2S on my Firebird
- I have coarsened the pitch on one of my 5030 props and did a thrust test - it had less static thrust as expected but may make it better when airborne (hopefully)
- I have cambered the wing leading edge so it is not a blunt shape creating drag and turbulence
- I can switch to 2S, but will need to upgrade some of the radio gear first - even though the receiver was advertised to handle up to 10v, I could not get it to work with a 2S battery
- I need to switch to 2S ultimately to get the best out of this model

Conclusions

- I was very disappointed I didn't manage to keep the plane in the air for more than 50m! I know how the Wright Brothers must have felt after their first 260m flight! Except, my plane didn't get wrecked...
- On the plus side, the plane didn't get damaged before I could get it to fly properly
- I have since made the mods to the prop and wing, and will be able to test it out again next week.
- I have also just ordered a new 2/3S receiver with built in brushless ESC, which should arrive in the next couple of weeks. This receiver will also allow me to configure flaperons, to enhance slow speed performance.
- This will hopefully unlock a lot of potential of this plane.

Usual sign-off.

Tags:

Magnum Reloaded, DMFV Jump Racer, Foam Build, Printing Foam, Indoor RC Plane, Indoor Aviation, Scratch Build, Build from plan, 4 Channel RC Plane, Easy Foam Build, 3D Stunt Plane, Maiden flight

Title

DMFV Jump Racer Maiden Flights #davesfunrc

Description

Check out my maiden flights of the recently built 50cm wingspan DMFV Jump Racer.

Plane Stats:

Wingspan: 56cm (22")

Length: 45cm (18")

Flying Weight (with 500mAh battery): 124g (4.37oz)

Static Thrust: 45g (250mAh battery), 55g (500mAh battery), 62g (800mAh battery)

Motor: Micro 1104 HM Brushless Motor 1-2S 4300KV Mini RC Quadcopter Brushless

Links

DMFV Jump Racer Build Video

DMFV Academy

ZART Shop in Melbourne

Link to receiver that didn't work with 2S

Link to 5ch 2/3S Receiver

Links

Magnum Reloaded Build Video:

<https://youtu.be/BpuUfvdKSmU>

DMFV Academy:

https://www.google.com/search?q=DMFV-Jump-Racer-Konstruktion&oq=DMFV-Jump-Racer-Konstruktion&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCDEyMzRqMGo3qAIA&sourceid=chrome&ie=UTF-8#fpstate=ive&vld=cid:d6b75a65,vid:Bi0jUbTTliU,st:0

ZART Shop in Melbourne:

ZART.COM.AU

Build

I started by printing out the plan using the tile option.

I scaled it for a 50cm wingspan (90%) of the original Jump Racer.

I printed the PDF plan 72% Poster to my printer (9 tiles).

<Voice over timelapse:>

I taped the sheets together and began pinning out the parts onto A3 and A4 sheets of printing foam.

The fuselage sections were intended to be folded. Printing foam is brittle, so I decided to make two 45° angle cuts and glue them together with PVA glue.

Videos:

- Glued two wings tougher
- Laminated the undercarriage
- Fuselage trial fit
- Wing section carbon fibre
- Paint scheme
- 50g so far plus 27g radio battery 13g
- Elevator and rudder moving surfaces v

<Add voice over>

I glued the wing and tail sections to fuselage.

I added a tail skid and attached the undercarriage

Foam parts weight 61g so far

Radio gear installation:

Weight with motor and radio gear came to 117g

But wait...I need wheels for taxiing - half the fun of indoor flying is taxiing and taking off!

And, I need a pilot - Snoopy!

With all this, the final weight came to 124g.

<Taxi test video>

We are all set for a maiden flight next week at indoor aviation!

Conclusions

Here are the Plane Stats:

Wingspan: 56cm (22")

Length: 45cm (18")

Flying Weight (with 500mAh battery): 124g (4.37oz)

Static Thrust: 45g (250mAh battery), 55g (500mAh battery), 62g (800mAh battery)

If my thrust test rig is accurate, I think it will fly OK indoors.

It won't however be able to do 3D flying outside.

I may need to reposer it at the later stage, if I want to do this.

Stay tuned for my next video, where I will do the maiden flights indoors.

Usual signify...

DMFV Plan

Tags:

Magnum Reloaded, DMFV Jump Racer, Foam Build, Printing Foam, Indoor RC Plane, Indoor Aviation, Scratch Build, Build from plan, 4 Channel RC Plane, Easy Foam Build, 3D Stunt Plane

Title

DMFV Jump Racer Scratch Foam Build #davesfunrc

Description

Join me on this exciting little project to build a 58cm DMFV Jump Racer from 3mm printing foam. Ever heard of this plane, well the DMFV Academy in Germany use this plan to teach children how to build a plane from flexi foam.

Alex Herera, one of my viewers, requested that I build this plane. I decided to build a 58cm version, using A3 and A4 sheets of 3mm Printing Foam from the ZART shop in Melbourne. Let's see how this build went!

Links

Magnum Reloaded Build Video

DMFV Academy

ZART Shop in Melbourne

Specs for the motor:

3.7v 63g 5030

